HealthRAG: Knowledge-Driven Medical QA System

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Abstract

Providing accurate and accessible medical information to non-experts in Vietnam is a pressing challenge. This thesis presents a Retrieval-Augmented Generation (RAG) system for medical question answering, leveraging 610 disease-related articles from youmed.vn. The system integrates three specialized retrieval methods: dense retrieval with a fine-tuned hiieu/halong_embedding model, BM25-based keyword search, and Knowledge Graph (KG)-based retrieval, all evaluated on a diverse QA dataset of over 5,000 questions. At 128 dimensions, dense retrieval achieves post-fine-tuning performance of Accuracy@3 of 99.35%, Recall@3 of 99.35%, and NDCG@3 of approximately 0.916, compared to 79.94%, 79.94%, and 0.735 pre-fine-tuning, reflecting improvements of 19.4% and 18.1%. BM25 excels in keyword precision but lacks semantic depth, while KG retrieval provides structured relational insights, limited by entity extraction quality. Integrating these methods via Reciprocal Rank Fusion combines semantic richness, keyword accuracy, and structural knowledge, enhancing retrieval effectiveness and answer quality. Google Gemini, as the Large Language Model, delivers precise, context-aware responses. This system establishes a robust Vietnamese medical Q&A framework, advancing healthcare information access in low-resource language settings with significant clinical and educational potential.

Keywords: RAG, medical Q&A, Embedding Model, KG, keyword search.

Table of Contents

1. Introduction 9
    1.1. Motivation 9
    1.2. Literature Review 10
    1.3. Related Work 11
        1.3.1. LLMs in Healthcare 11
        1.3.2. RAG Systems 12
        1.3.3. RAG in Medical Question Answering 13
        1.3.4. Text Embedding Models for Vietnamese 14
        1.3.5. Keyword-Based Search 16
        1.3.6. KG in Information Retrieval 16
    1.4. Objectives 17
2. Methodology 18
    2.1. System Architecture Overview 18
    2.2. Data Collection and Preprocessing 19
        2.2.1. Medical Article Corpus from youmed.vn 19
        2.2.2. Data Preparation for Dense Retrieval 21
        2.2.3. Data Preparation for Sparse Retrieval 22
        2.2.4. Data Preparation for KG 22
    2.3. Halong Embedding Model 23
        2.3.1. Halong Embedding Pre-training 23
        2.3.2. Fine-tuning on Medical QA Dataset 24
    2.4. Retrieval Strategy 25
        2.4.1. Embedding-based Search 25
        2.4.2. Keyword-based Search 26
    2.5. KG Construction and Integration 26
        2.5.1. Entity and Relation Extraction 26
        2.5.2. KG Utilization in Retrieval and Generation 27
    2.6. Gemini LLM for Answer Generation 27
        2.6.1. Prompt Engineering 27
        2.6.2. Post-processing and Safety Considerations 28
3. Experiment 28
    3.1. Datasets & Evaluation Metrics 28
        3.1.1. Datasets 28
        3.1.2. Evaluation Metrics 29
    3.2. Results & Analysis 30
        3.2.1. Halong Embedding Performance 30
        3.2.2. Keyword Search Performance (BM25) 32
        3.2.3. KG Evaluation 33

4. Conclusion & Future Work 34
References 35

List of Tables

Table 1. Dataset Statistics 29
Table 2. Comparison of Retrieval Metrics for Halong Embedding (Dim=128) Before and After Fine-Tuning 31
Table 3. Retrieval Metrics for BM25 Keyword Search 32
Table 4. Link Prediction Evaluation Performance 34

List of Figures

Figure 1. LLM Hallucination in Medical Contexts 12
Figure 2. Basic RAG System Architecture 13
Figure 3. Challenges for RAG in Medical Domain 14
Figure 4. Semantic Space of Text Embeddings and Cosine Similarity 15
Figure 5. Example of a Medical KG Snippet 17
Figure 6. Overall Pipeline of the Vietnamese Medical RAG System 18
Figure 7. Distribution of Article Lengths in the Medical Corpus 20
Figure 8. Top 20 Most Frequent Medical Article Keywords in the Corpus 20
Figure 9. Number of Images Associated with Top 20 Articles 21
Figure 10. General Architecture of a Transformer Encoder 23

List of Abbreviations and Acronyms

AI: Artificial Intelligence
API: Application Programming Interface
ATLAS: Anatomical Tracings Of Lesions After Stroke
BM25: Best Match 25
BERT: Bidirectional Encoder Representations from Transformers
GPU: Graphics Processing Unit
FAISS: Facebook AI Similarity Search
KG: Knowledge Graph
LLM: Large Language Model
MAP: Mean Average Precision
MNRL: Multiple Negative Ranking Loss
MRR: Mean Reciprocal Rank
MR: Mean Rank
NDCG: Normalized Discounted Cumulative Gain
NER: Named Entity Recognition
NLP: Natural Language Processing
MIMIC: Medical Information Mart for Intensive Care
IDs: infectious diseases
PaLM: Pathways Language Model
REALM: Rapid Estimate of Adult Literacy in Medicine
QA: Question Answering
RAG: Retrieval-Augmented Generation
RDF: Resource Description Framework
RE: Relation Extraction
TF-IDF: Term Frequency-Inverse Document Frequency
GPT: Generative Pre-trained Transformer
UMLS: Unified Medical Language System
FiD: Fusion-in-Decoder
KGE: Knowledge Graph Extraction
USMLE: United States Medical Licensing Examination

1. Introduction

1.1. Motivation

In the contemporary digital age, the internet has emerged as a primary resource for individuals seeking health-related information. This accessibility, however, often comes with significant challenges, as users navigate an overwhelming volume of data, encounter varying levels of accuracy, and struggle with the complexity of medical jargon, making it difficult to discern reliable and comprehensible information. Studies have shown that a substantial majority of internet users, often exceeding 75% in many regions, actively search for health information online. Despite this high demand, a significant portion of these users express concerns regarding the trustworthiness and clarity of digital health sources, with some assessments indicating that fewer than half of individuals fully trust information found on social media platforms for health advice. Misinformation or delayed access to critical health insights can have serious consequences, impacting personal well-being, influencing healthcare decisions, and even jeopardizing patient safety. While traditional search engines frequently return a multitude of irrelevant or overly technical results, compelling users to sift through extensive content, LLMs, despite their impressive generative capabilities, pose a significant risk of "hallucination"—generating factually incorrect or unsupported information, a critical drawback in sensitive domains like healthcare. This thesis is therefore motivated by the urgent need to bridge the gap between complex medical literature and the public's demand for accessible, accurate health information. We aim to develop a robust Vietnamese QA system specifically tailored for medical diseases, leveraging a meticulously curated corpus of 610 articles from youmed.vn, a reputable local health portal. Our project seeks to provide a reliable, efficient, and user-friendly platform for citizens to query and receive precise answers, thereby enhancing public health literacy and supporting informed healthcare decisions in Vietnam.

1.2. Literature Review

The field of QA in the medical domain has undergone significant evolution in recent years, driven by advances in deep learning, natural language understanding, and the emergence of LLMs. Traditional QA systems relied heavily on rule-based approaches or shallow pattern-matching techniques, which offered limited flexibility in handling natural, unstructured queries. With the advent of transformer-based models, particularly BERT and its domain-specific variants like BioBERT and ClinicalBERT, medical QA systems began to benefit from context-aware language representations. These models demonstrated improvements in named entity recognition, relation extraction, and reading comprehension within clinical texts [1].

However, transformer-based models also introduced challenges, particularly in maintaining factual consistency during generation. Studies such as Ji et al. [2] and Singhal et al. [3] have shown that even the most advanced LLMs, including GPT-4 and PaLM-2, may produce factually inaccurate or unsupported responses—a phenomenon commonly referred to as "hallucination." In high-stakes fields like healthcare, this limitation is not merely a technical flaw but a significant barrier to safe adoption. Consequently, the research community has increasingly shifted toward RAG, a hybrid approach that integrates the retrieval of grounded knowledge with generative modeling. RAG systems have been shown to reduce hallucinations, enhance factual accuracy, and support explainable AI in medical settings [4][5].

The RAG paradigm was introduced by Lewis et al. [6], where a retriever selects relevant documents and a generator uses them to produce answers. This architecture has since been refined in systems like FiD [7], ATLAS [8], and REALM [9], which improve how retrieved context is integrated during generation. In the biomedical domain, RAG-based systems such as BioRAG [10] and MedRAG [11] have incorporated curated corpora from PubMed or UMLS to support evidence-based QA. These systems outperform end-to-end generative models on tasks requiring accurate recall of medical facts or multi-hop reasoning over structured knowledge.

Parallel to model architecture, retrieval quality has emerged as a critical research focus. Sparse retrieval techniques like BM25 offer high precision for keyword-dense queries [12], while dense retrievers like Sentence-BERT [13] and E5 [14] leverage semantic embeddings to handle paraphrasing and indirect expressions. More recently, hybrid retrieval methods have been

proposed, combining both paradigms to improve robustness across varying query types [15]. Additionally, the incorporation of KG into QA systems has opened new opportunities for structured reasoning and explainable inference. Medical KGs like UMLS and SNOMED-CT support query expansion and enhance retrieval through explicit modeling of relationships between symptoms, diseases, and treatments [16].

In the context of Vietnamese, QA remains an underexplored challenge due to limited data resources, lack of pretrained domain models, and language-specific complexities such as tones and compound segmentation. While some progress has been made with Vietnamese embeddings like PhoBERT and ViHealthBERT [17], most research still centers on general NLP tasks rather than domain-specific applications like healthcare. This thesis builds upon these foundational efforts by introducing a fully localized Vietnamese medical RAG system—combining domain-adapted embedding (Halong Embedding), dense and sparse retrieval, and a structured Vietnamese medical KG. Through this integrated framework, we aim to advance the state of QA in Vietnamese and provide a practical, explainable solution for delivering accurate health information to the public.

1.3. Related Work

1.3.1. LLMs in Healthcare

The rise of LLMs, such as GPT-4 [18] and PaLM-2 [19], has significantly advanced natural language processing across various domains, including healthcare. These models demonstrate strong capabilities in clinical summarization, medical QA, triage assistance, and the generation of patient education content. Despite their fluency and broad knowledge, LLMs remain prone to hallucination—the generation of plausible but factually inaccurate information. In high-stakes fields like medicine, such hallucinations present serious challenges for safe and trustworthy deployment [20]. Recent evaluations, including MedQA-USMLE [21] and MultiMedQA [22], have assessed LLMs' medical proficiency. While models like GPT-4 can pass USMLE-style examinations, they often generate unsupported statements, particularly in the absence of external knowledge grounding. Consequently, a growing number of QA systems have adopted retrieval-augmented approaches to ensure factual accuracy.

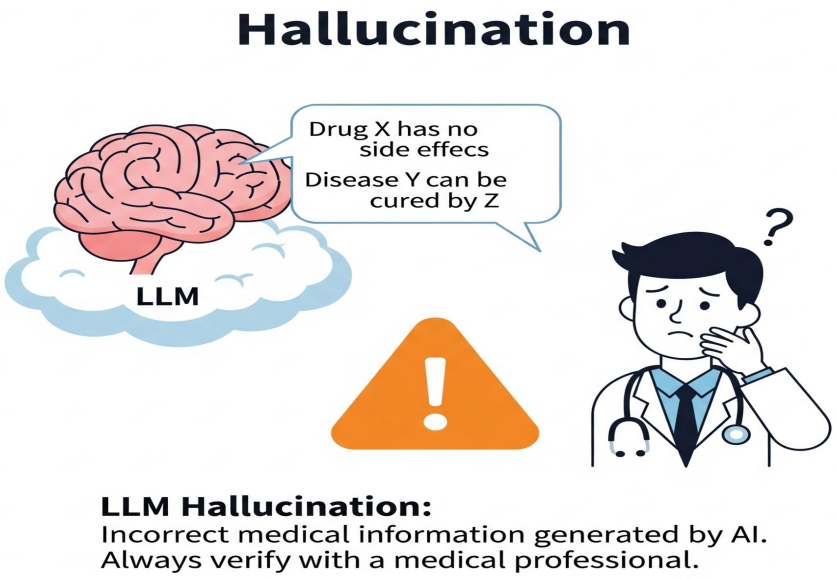


Figure 1: LLM Hallucination in Medical Contexts

In our system, we adopt Gemini [23], Google's most advanced LLM, designed for long-context and multimodal reasoning. Gemini exhibits improved factual alignment over its predecessors and is inherently suited for integration within retrieval-augmented pipelines. When paired with a robust Vietnamese retrieval module, Gemini enables contextually grounded generation, effectively minimizing hallucinations while maintaining fluency and relevance in the medical domain.

1.3.2. RAG Systems

RAG introduced by Lewis et al. [6], is a prominent framework designed to enhance factual accuracy by integrating external knowledge sources into the generative process. Rather than relying solely on a language model's internal memory, RAG retrieves relevant documents from a knowledge corpus and uses them as context to condition the output. This approach has proven especially effective in open-domain and knowledge-intensive tasks. The original RAG architecture comprises two core components: a bi-encoder retriever and a sequence-to-sequence generator. Subsequent improvements, such as FiD [8], enable the decoder to attend to multiple retrieved passages. Other variants include REALM [9], which jointly trains retriever and generator, and ATLAS [10] retrieval-aware fine-tuning.

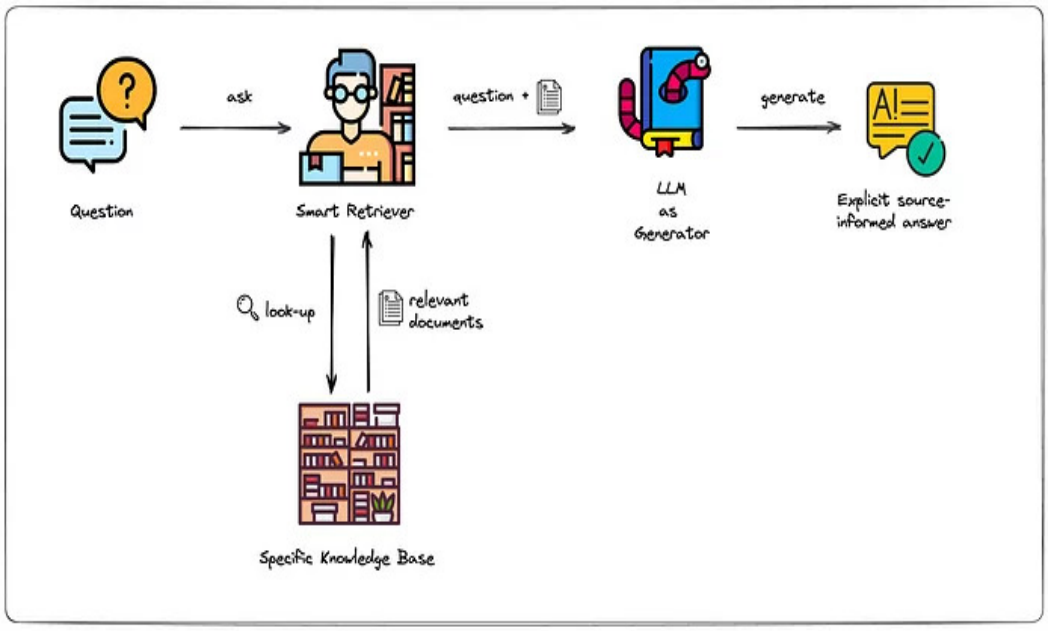


Figure 2: Basic RAG System

1.3.3. RAG in Medical Question Answering

In our system, we adopt a modular RAG design tailored for Vietnamese medical QA. Halong Embedding serves as the dense retriever, while Gemini functions as the generator. This separation enables continuous updates to the retriever corpus without re-training the generator, a key advantage in domains where information evolves rapidly.

Biomedical QA is a domain where factual precision and knowledge grounding are critical, making it an ideal use case for RAG. Notable systems such as BioRAG [11] and MedRAG [12] have demonstrated the effectiveness of combining domain-specific retrieval mechanisms with large generative models. These architectures typically utilize curated biomedical corpora—such as PubMed, UMLS, or MIMIC-III—to provide verifiable, evidence-based context for generated responses.

Although these models perform well in English-language settings, their applicability to other languages remains limited. Most biomedical RAG systems have been developed and fine-tuned exclusively on English data,

with few attempts to adapt them to multilingual contexts. Existing multilingual extensions often underperform due to a lack of high-quality training resources and domain-specific tuning for target languages.

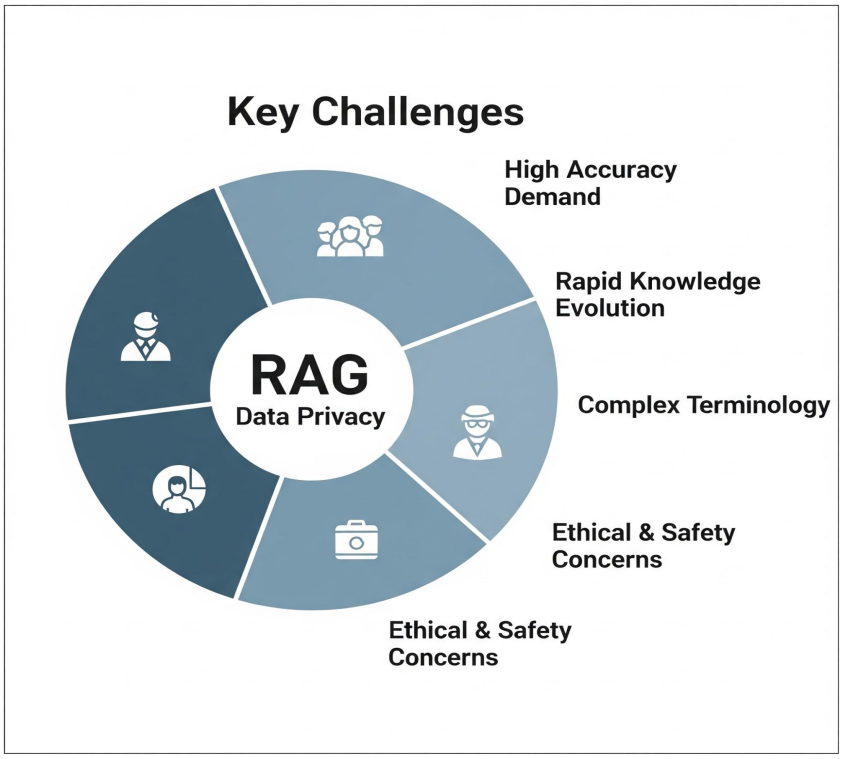


Figure 3: Key Challenges for RAG in Medical Domain

1.3.4. Text Embedding Models for Vietnamese

To address this gap, our work introduces a Vietnamese-specific medical RAG pipeline. It leverages a curated corpus of over 610 Vietnamese medical articles and a QA dataset consisting of more than 5000 annotated examples. We fine-tune the retriever component using Halong Embedding to capture local medical semantics and construct a Vietnamese medical KG to support structured retrieval. This combination allows us to build a system capable of retrieval-grounded generation in a low-resource, high-stakes domain—minimizing hallucination risks while ensuring linguistic and clinical relevance.

Semantic retrieval is a foundational pillar of any RAG system. Dense retrievers rely on text embedding models to project both queries and

candidate documents into a shared high-dimensional vector space, enabling efficient similarity comparison using metrics such as cosine similarity. Early embedding approaches, such as Word2Vec and GloVe, introduced static vector representations, while more recent models like Sentence-BERT [13], E5 [14], and MPNet [24] have advanced the field by producing context-aware, sentence-level embeddings that capture rich semantic relationships.

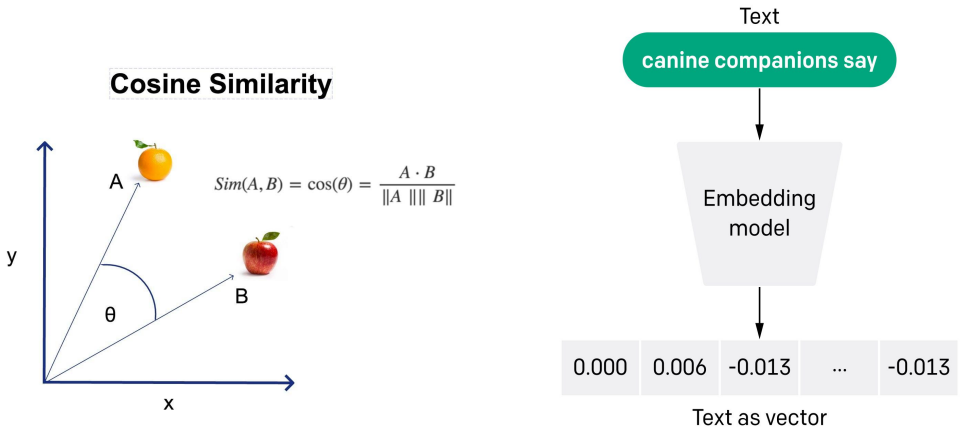


Figure 4: Semantic Space of Text Embeddings and Cosine Similarity

Despite these advancements, embedding models tailored to Vietnamese remain relatively underdeveloped. The Vietnamese language presents unique linguistic challenges—including tonal variation, compound morphology, and inconsistent word segmentation—that limit the effectiveness of generic multilingual or transfer-learned models. This poses a significant barrier for domain-specific tasks such as medical QA.

To address this, we utilize Halong Embedding, a Vietnamese-specific text embedding model adapted for semantic retrieval in the healthcare domain. The model was pre-trained on over 100,000 Vietnamese QA pairs, enabling it to learn specialized representations that align closely with medical terminology and local expression. In our system, Halong Embedding functions as the dense retriever, enabling accurate semantic matching between user queries and relevant documents. It has also been fine-tuned in our experiments to improve alignment with Vietnamese clinical language, effectively capturing nuanced medical relationships—such as recognizing that *viêm amidan* (tonsillitis) is contextually related conditions.

1.3.5. Keyword-Based Search

Although dense embedding models excel at capturing semantic similarity, keyword-based retrieval remains a powerful tool, especially in domains like medicine, where terminology is highly specific and standardized. BM25 [16], a probabilistic ranking function based on term frequency and inverse document frequency, continues to be a strong baseline for exact term matching in information retrieval tasks. In the context of medical QA, BM25 proves particularly effective for queries containing precise keywords such as disease names, drug classes, or medical codes. Unlike semantic models, it does not rely on contextual understanding, which allows it to retrieve relevant documents even when the input includes rare

In our system, BM25 is used in tandem with Halong Embedding to form a hybrid retrieval setup. This combination enhances both recall and precision by leveraging the lexical accuracy of BM25 and the contextual depth of dense embeddings. It ensures that critical documents are not missed due to synonym mismatch or vocabulary variation, making the retrieval process more robust for Vietnamese medical QA.

1.3.6. KG in Information Retrieval

KG offers a structured way to represent domain knowledge by modeling entities—such as diseases, symptoms, and treatments—and their relationships (e.g., causes, treated by) in the form of subject–predicate–object triples. In the biomedical domain, large-scale KGs like UMLS, SNOMED-CT, and DrugBank have been extensively used to support clinical decision-making, semantic search, and explainable QA [25]. KG enables multi-hop reasoning, term disambiguation, and fact verification. For instance, a KG can connect "viem phoi" (pneumonia) → "vi khuan" (bacteria) → "khang sinh" (antibiotics), allowing inference of treatment paths not explicitly stated in a single document. This structure is especially valuable in high-stakes domains where explainability and correctness are critical.

In our system, we construct a Vietnamese medical KG by extracting disease–symptom–treatment relations from our article corpus. The KG is integrated into both retrieval and generation pipelines, guiding Gemini toward medically grounded concepts. It functions as both a semantic enrichment tool and a factual grounding layer within our hybrid architecture.



Figure 5: Example of a Medical KG Snippet

1.4. Objectives

The objective of this thesis is to design and implement a RAG system tailored to the Vietnamese medical domain, with the goal of generating accurate, trustworthy, and contextually relevant answers to user queries. To achieve this, the system integrates a dense retriever (Halong Embedding), a LLM (Gemini), and a structured KG representing disease–symptom–treatment relationships. A curated corpus of over 600 Vietnamese medical articles and 5,000+ QA pairs is used to fine-tune and evaluate system components. The research also explores hybrid retrieval strategies that combine semantic, keyword-based, and graph-based signals to improve both precision and recall. Through this approach, the thesis aims to address the challenge of factual inconsistency in generative models and contribute a robust, extensible framework for Vietnamese healthcare QA.

2. Methodology

2.1. System Architecture Overview

The proposed Vietnamese medical QA system operates as an end-to-end RAG framework, meticulously designed for accurate and contextually relevant medical information delivery. The process begins with query preprocessing, followed by a Hybrid Retrieval Module that synergistically combines dense retrieval (using Halong Embedding from a Vector Database) and sparse retrieval (keyword-based indexing). This module also integrates data from a Vietnamese Medical KG and identifies relevant image captions. All retrieved information is then subjected to context aggregation before being fed into the Generative Module, powered by LLM. This ensures the synthesized answer is factually grounded and mitigates hallucination. Finally, the generated textual response undergoes post-processing and multimodal integration with relevant images, culminating in a comprehensive output for the user.

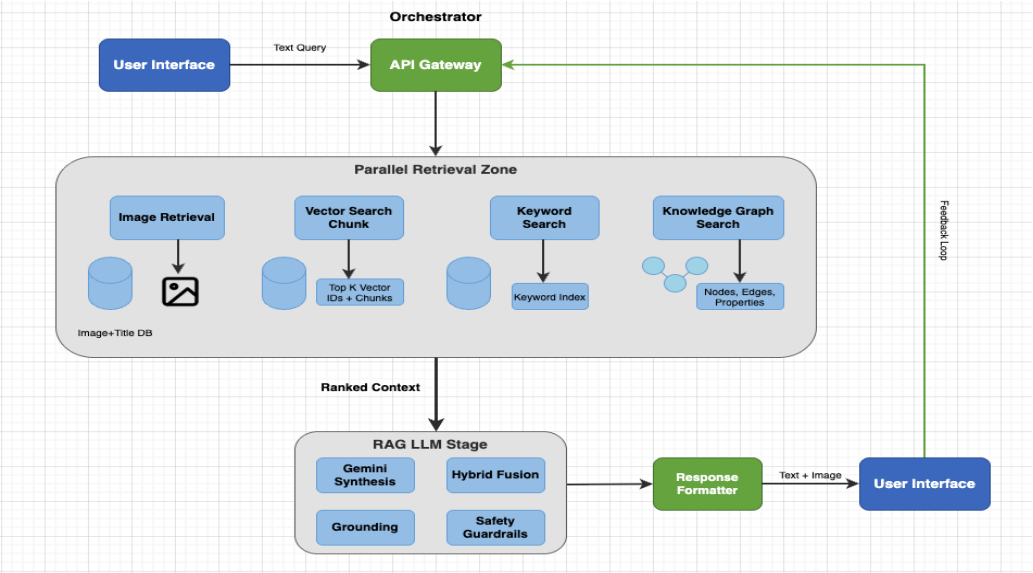


Figure 6: Overall Pipeline of the Vietnamese Medical RAG System

2.2. Data Collection and Preprocessing

2.2.1. Medical Article Corpus from youmed.vn

The efficacy and robustness of a RAG system are fundamentally dependent on the quality, relevance, and comprehensive nature of its underlying knowledge base. For this Vietnamese medical RAG system, a dedicated corpus of 610 medical articles was meticulously constructed and preprocessed, sourced from youmed.vn. This platform was chosen for its extensive collection of healthcare information presented in Vietnamese, making it an invaluable resource for our domain-specific needs.

The data acquisition process leveraged automated web scraping techniques. Custom Python scripts were developed to systematically navigate the youmed.vn website, extracting core textual content from individual article pages, including main article bodies, subheadings, and lists. Simultaneously, associated image URLs were identified, and 1882 images were downloaded and then organized into disease-specific folders. This structured storage (e.g., images for "Addison" disease in an 'Addison' folder) implicitly links visual content to its corresponding textual article, assuming the folder name matches the article's primary topic.

Following the acquisition, the raw web-scraped data underwent a rigorous cleaning and preprocessing pipeline to ensure its suitability for downstream NLP and retrieval tasks. This pipeline encompassed several critical steps: HTML tag and script removal to isolate pure textual information; irrelevant content filtering to eliminate advertisements, navigation menus, and other extraneous web elements; character normalization to standardize Vietnamese diacritics and character encodings for consistency; and whitespace and redundancy elimination to standardize spacing and remove clutter. This comprehensive cleaning process transformed the heterogeneous raw data into a clean, consistent, and structured multimodal corpus. To gain a comprehensive understanding of the characteristics, distribution, and potential patterns within this processed medical corpus, various data visualization techniques were employed. The initial analysis involved visualizing the distribution of article lengths. This histogram, as depicted in Figure 7, illustrates the frequency of different article lengths, measured by word count, providing insights into the typical size and overall uniformity of the textual documents. This information is crucial for determining optimal chunking strategies later in the RAG pipeline.

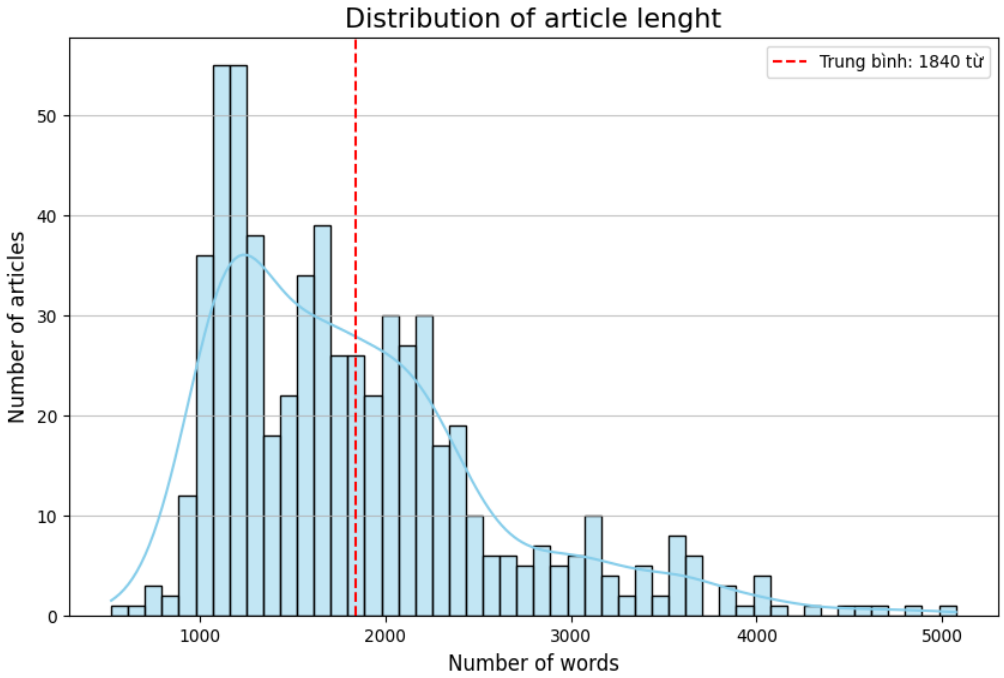


Figure 7: Distribution of Article Lengths in the Medical Corpus

Further analysis involved examining keyword frequency within the textual corpus. Figure 8 presents a bar chart highlighting the 20 most commonly occurring words, revealing the dominant terminology and core themes.

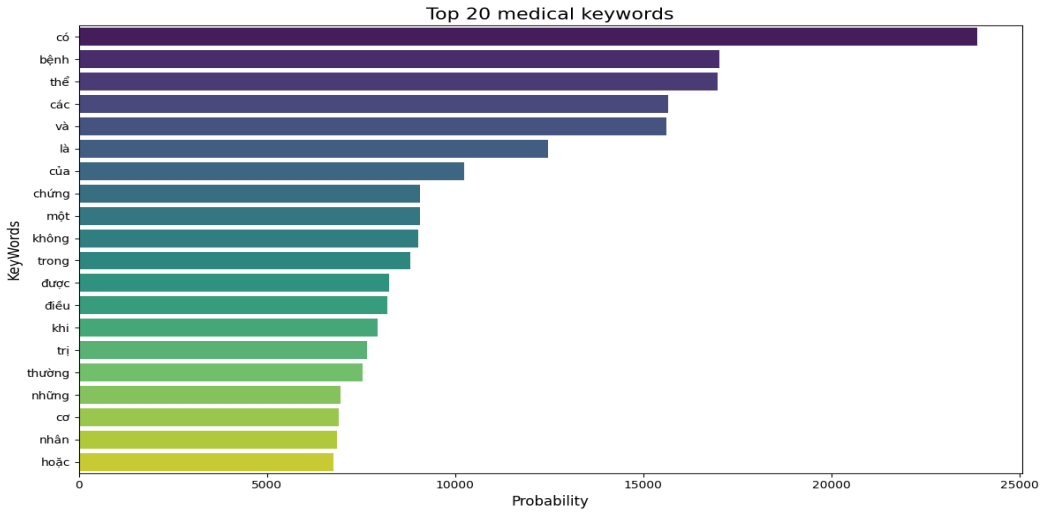


Figure 8: Top 20 Most Frequent Medical Article Keywords in the Corpus

To assess the multimodal richness of the corpus, the distribution of image data per article was analyzed. Figure 9 displays a bar chart showing the number of images linked to the top 20 articles with the most associated visuals. This illustrates the extent to which each medical article (or disease topic, derived from the folder name) is supplemented by associated image data, highlighting the multimodal support across different sections of the knowledge base.

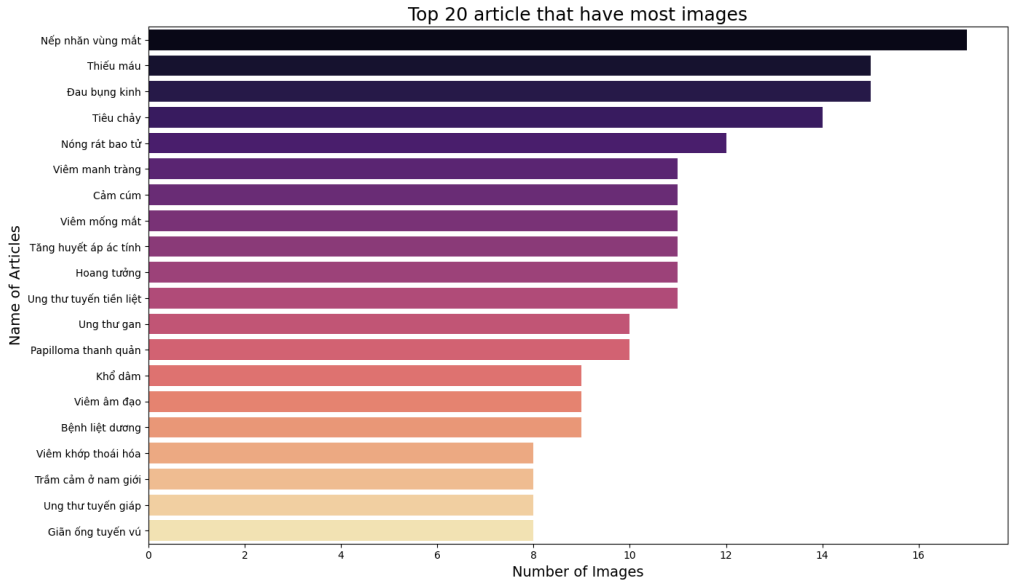


Figure 9: Number of Images Associated with Top 20

2.2.2. Data Preparation for Dense Retrieval

For dense retrieval, the 610 cleaned medical articles are first segmented into smaller, overlapping chunks, each 200 words with a 50-word overlap, to optimize context and retrieval granularity. Each chunk is then transformed into a high-dimensional vector embedding using a customized Halong Embedding model after finetuning , a pre-trained Vietnamese Sentence Transformer. These semantic embeddings are subsequently indexed within FAISS (Facebook AI Similarity Search) using an IndexFlatL2 for rapid nearest neighbor queries. Essential metadata, including original article and chunk IDs, along with the full chunk text, is stored separately to facilitate mapping retrieved indices back to their original content, a process seamlessly integrated with frameworks like LangChain in the RAG system.

2.2.3. Data Preparation for Sparse Retrieval

For sparse retrieval, the 610 cleaned medical articles are segmented into paragraph-level passages to enhance lexical match precision. Each passage is preprocessed by removing Vietnamese stopwords and normalized for consistency. The passages are then indexed using the BM25 algorithm via the rank-bm25 library, which computes relevance scores based on term frequency, inverse document frequency, and document length normalization. At retrieval time, user queries are processed similarly and matched against the indexed passages using BM25 scoring. Retrieved passage metadata, including article and paragraph IDs, is stored separately to support integration with RAG frameworks.

2.2.4. Data Preparation for KG

To complement dense and sparse retrieval components in the Vietnamese Medical RAG system, a domain-specific KG was constructed from the textual and visual data. This structured representation of medical entities and relationships enhances semantic retrieval and contextual grounding. The preparation pipeline for the KG. Entity Extraction: Medical entities such as diseases, symptoms, treatments, diagnostic procedures, and medications are automatically identified from the article corpus using a custom Vietnamese medical Named Entity Recognition (NER) model. This model is either fine-tuned on domain-specific datasets or uses pre-trained Vietnamese biomedical NERs (e.g., PhoNER-COVID). Relation Extraction: Relationships between entities are determined using either rule-based patterns or deep learning-based relation extraction techniques. Typical relations include "treated by", "symptom of", "diagnosed by", "associated with", and "described in" (for images). Entity Normalization: Extracted entities are normalized against standardized vocabularies such as UMLS or SNOMED CT, where possible, and linked to canonical Vietnamese labels to reduce redundancy and improve consistency across the graph. Graph Construction: Extracted triples (subject, predicate, object) are converted into a structured graph using RDF or stored in graph databases such as Neo4j. Each node represents a medical entity, and each edge encodes a semantic relationship.

Integration with Retrieval Module: The KG supports hybrid retrieval by providing structured relational data that complements unstructured text and image embeddings. When a user poses a question, the system can retrieve relevant subgraphs, helping to ground the LLM's generation in

verified medical facts and reducing hallucination. This graph-centric data representation not only supports enhanced retrieval accuracy but also facilitates multimodal reasoning, particularly when combining textual and visual evidence in the answer synthesis process.

2.3. Halong Embedding Model

2.3.1. Halong Embedding Pre-training

The Halong Embedding model is a Sentence Transformer model specifically designed for Vietnamese text embedding. It is built upon the robust intfloat/multilingual-e5-base model, serving as its foundational pre-trained base. The multilingual-e5-base model itself is a powerful Transformer-based encoder capable of mapping sentences and paragraphs to a 768-dimensional dense vector space, with a maximum sequence length of 512 tokens, making it suitable for a wide range of multilingual NLP tasks.

The initial adaptation phase for Halong Embedding leveraged this strong multilingual base. This step ensures the model inherits robust general language understanding capabilities before being specialized for the medical domain. The underlying Transformer encoder architecture allows it to effectively process input sequences, capturing long-range dependencies within the text to produce contextualized embeddings.

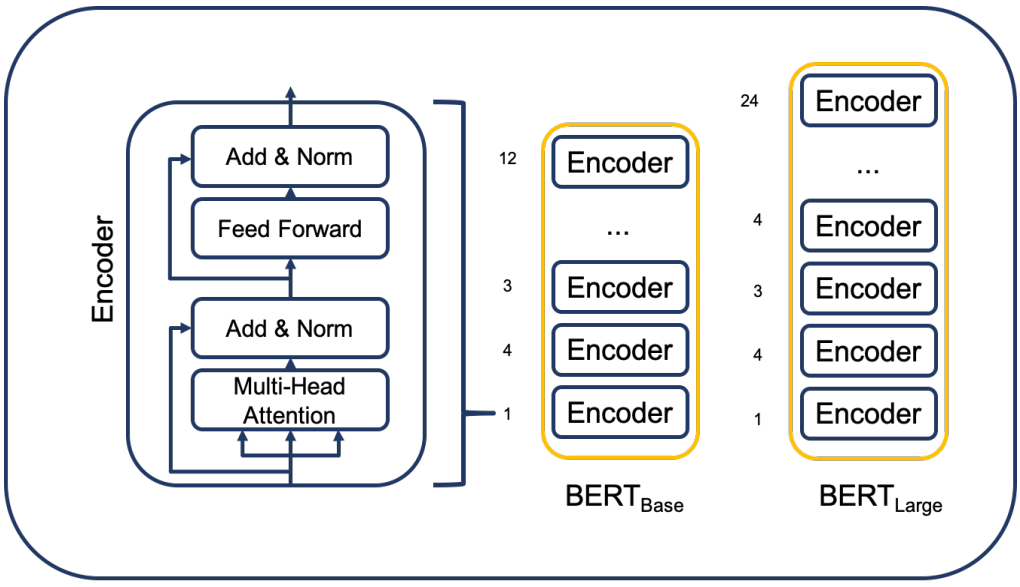


Figure 11: General Architecture of a Transformer Encoder

2.3.2. Fine-tuning on Medical QA Dataset

Following its base adaptation, the Halong Embedding model undergoes a crucial fine-tuning phase to specialize its semantic understanding for the medical domain and optimize for production efficiency, including flexible embedding dimensionality. This fine-tuning uses a Siamese network architecture on an in-house dataset of approximately 100,000 questions and related document examples. The objective is to learn an embedding space where semantically similar medical questions and answer chunks are mapped closely, while dissimilar pairs are pushed apart. To achieve this, the fine-tuning process leverages a composite loss function that combines Multiple Negative Ranking Loss and a specialized Matryoshka Loss.

MNR Loss is a loss function used in contrastive learning and information retrieval. It aims to train a model to rank relevant items higher than irrelevant ones. MNR Loss works by comparing the similarity between a query and relevant and irrelevant documents. It encourages the model to assign high similarity scores to query-relevant document pairs and low similarity scores to query-irrelevant document pairs.

$$\mathcal{L}_{MNR} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp\left(\frac{sim(q_i,d_i^+)}{\tau}\right)}{\sum_{j=1}^{N}\exp\left(\frac{sim(q_i,d_j)}{\tau}\right)} \tag{1}$$

Where sim is cosine similarity, q_i is the query vector, d_i^+ is the relevant document corresponding to q_i, d_j represents all documents in the batch, N is the batch size, and τ is a temperature parameter controlling the softness of the distribution.

Matryoshka Loss is a technique used to train embedding models such that they produce vector representations that can be easily reduced in size without losing too much information. It is inspired by Russian Matryoshka dolls, in which smaller dolls are nested inside larger ones.

$$\mathcal{L}_{LM} = \sum_{i=1}^{k} w_i \cdot \mathcal{L}(emb_{[:d_i]}) \tag{2}$$

Where emb is the full-size embedding vector, $emb[:\ d_i]$ represents the embedding vector truncated to dimension d_i, w_i is the weight assigned to the loss of the truncated embedding, $\mathcal{L}(\cdot)$ is the base loss function (e.g.,

contrastive or MNR loss) applied to the truncated embeddings, and k is the total number of truncation levels considered. This formulation encourages the model to produce embeddings that retain important information even when truncated, enabling flexible and efficient representations for downstream tasks.

2.4. Retrieval Strategy

The effectiveness of our RAG system hinges on its ability to intelligently retrieve relevant information. We employ a multi-faceted retrieval strategy that combines semantic understanding with precise keyword matching. This approach integrates embedding-based search, keyword-based search, and a hybrid mechanism to ensure comprehensive and accurate information retrieval for any given user query.

2.4.1. Embedding-based Search

Embedding-based search, also known as dense retrieval, forms the semantic backbone of our system. It leverages the Halong Embedding model (detailed in Section 2.3) to transform a user's query into a 768-dimensional dense vector, effectively capturing its underlying meaning. This query embedding is then used to perform a rapid Cosine Similarity search against the FAISS index , which contains pre-computed embeddings of all medical article chunks. The process efficiently identifies and returns a ranked list of the Top-K most semantically similar document chunks. This method excels at comprehending nuanced queries, handling synonyms, and retrieving contextually relevant information even when exact keywords aren't present.

$$similarity(q, d_i) = \frac{q \cdot d_i}{\|q\| \, \|d_i\|} \tag{3}$$

Where q is the query vector, d_i is the document vector, $q \cdot d_i$ denotes the dot product between q and d_i, and $\| \cdot \|$ represents the Euclidean norm of a vector. This measure evaluates the cosine of the angle between the two vectors, providing a normalized similarity score between -1 and 1, with higher values indicating more similar vectors.

2.4.2. Keyword-based Search

Keyword-based search, or sparse retrieval, complements the embedding approach by focusing on direct lexical matches. This method is highly effective for queries containing specific or unique terms and relies on an inverted index . The user's query is tokenized and preprocessed, then used to query this inverted index. Retrieved chunks are subsequently ranked using the BM25 algorithm, which evaluates relevance based on factors like term frequency, inverse document frequency, and document length. This process yields a ranked list of the Top-M most lexically relevant chunks, ensuring high precision for specific keyword occurrences.

$$BM25(q, D) = \sum_{i=1}^{|q|} IDF(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{avgdl}\right)} \tag{4}$$

Where q_i is the i-th term in the query, $|q|$ is the number of terms in the query, $f(q_i, D)$ is the frequency of term q_i in document D, $|D|$ is the length of the document, $avgdl$ is the average document length across the collection, k_1 and b are hyperparameters controlling term frequency saturation and document length normalization, and $IDF(q_i)$ is the inverse document frequency of q_i, measuring its rarity across the document collection. This formula balances term frequency with document length and global term importance to produce a relevance score for ranking documents.

2.5. KG Construction and Integration

2.5.1. Entity and Relation Extraction

To build a structured and semantically meaningful representation of Vietnamese medical knowledge, entity and relation extraction were conducted on the cleaned article corpus. A custom Vietnamese medical Named Entity Recognition (NER) pipeline was applied to detect key domain-specific entities, including diseases, symptoms, treatments, drugs, causes, and affected organs. This NER module was developed using transfer learning with pre-trained Vietnamese language models (e.g., PhoBERT, ViHealthBERT) and further fine-tuned on curated biomedical datasets.

For relation extraction, both rule-based pattern matching and supervised learning models were used. Labeled datasets were annotated with relational triples such as: (dai thao duong, co trieu chung, Khat nuoc),(viem gan B,

dieu tri boi, tenofovir), (sot xuat huyet, chan doan bang, xet nghiem NS1). These were extracted using dependency parsing and semantic similarity scoring to ensure accuracy. The resulting triples formed the foundation of the KG structure.

2.5.2. KG Utilization in Retrieval and Generation

Once constructed, the KG serves as a structured external memory that enhances both the retrieval and generation phases of the RAG pipeline. In the Retrieval Stage, the KG is queried using SPARQL-like patterns or graph traversal to extract relevant subgraphs related to user questions. The retrieved subgraphs provide additional context that complements the output of dense and sparse retrievers. These subgraphs are serialized into text (e.g.,'Benh dai thao duong co trieu chung khat nuoc dieu tri bang Insulin') and concatenated with retrieved chunks for final input into the LLM. In the Generation Stage, the KG ensures factual grounding by providing explicit, curated relationships that constrain the generation of the LLM. It reduces hallucination by offering a structured reference that the model can anchor to, especially for rare or complex medical conditions. When multimodal generation is needed, the KG also provides links between entities and associated images or captions to enable visual grounding.

2.6. Gemini LLM for Answer Generation

2.6.1. Prompt Engineering

Prompt engineering is the meticulous process of crafting effective inputs to guide the LLM's behavior and optimize its performance in answer generation. In the RAG system, this is critical to ensure the Gemini LLM accurately leverages the provided context, minimizes hallucination, and adheres to specific output requirements.

System Instructional Prompt: This segment explicitly defines the LLM's persona, provides strict directives to adhere only to the given context, and outlines desired output formats or behaviors. Retrieved Context: This primary source of factual information comprises two distinct parts, both transformed into textual segments for seamless LLM processing. The retrieval process produces three distinct outputs: Embedding-based Chunks (Result A), which are the top-N document chunks selected through embedding similarity and presented in a clear, delimited format, BM25-based Chunks (Result B), which are the top-N chunks ranked by

keyword-driven BM25 scoring, and Transformed KG Facts (Result C), which are structured facts extracted from the Knowledge Graph. User Query: The original question posed by the user, appended to instruct the LLM on the specific information to extract or synthesize from the aggregated context.

2.6.2. Post-processing and Safety Considerations

After the LLM generates an answer, post-processing refines the output and ensures its safety, which is crucial in the medical field. This involves formatting refinements for readability and crucial safety checks. These checks include hallucination detection (identifying information not supported by context), filtering harmful or unsafe content, and adding a disclaimer (e.g., stating the information isn't professional medical advice). This entire process ensures the final output is reliable, safe, and helpful.

3. Experiment

This section outlines the experimental setup, datasets, evaluation metrics, and results for our RAG system designed for medical QA in Vietnamese. We evaluate the system by focusing on the retrieval performance of Halong Embedding, the answer generation quality using Gemini LLM, and the overall system effectiveness. Importantly, we conducted the evaluation on the entire dataset without splitting it into separate subsets, and the same dataset was used for both fine-tuning and evaluation, ensuring a consistent approach.

3.1. Datasets & Evaluation Metrics

3.1.1. Datasets

We constructed two datasets to support the development and evaluation of our RAG system: the Medical Article Corpus and the Combined Dataset. Medical Article Corpus: The corpus consists of 610 medical articles crawled from *youmed.vn*, with each article dedicated to a single disease, covering a total of 610 unique diseases. The articles were collected using a Python-based web scraper (BeautifulSoup). We filtered out irrelevant content such as advertisements, navigation menus, and duplicate sections, reducing noise by approximately 20%. The text was cleaned by correcting typos, removing special characters, and standardizing Vietnamese medical terminology (e.g., "viêm phi" for pneumonia, "ái tháo ng" for diabetes). The final corpus contains 1,122,536 words, with an average article length of approximately 1840 words (1,122,536 / 610).

Table 1. Dataset Statistics

Metric	Value
Number of Articles	610
Total Words	1,122,536
Number of QA Pairs	5215
Average Words per Article	1840

We manually created a QA dataset for both fine-tuning and evaluation of the RAG system. For each of the 610 articles, we crafted 8-9 questions related to symptoms, treatments, and general notes, resulting in a total of more than 5000 QA pairs. This entire dataset was used for both fine-tuning Halong Embedding and evaluating the system's performance, ensuring consistency across training and testing phases. The dataset was not split into separate subsets, allowing a holistic assessment of the system's capabilities.

3.1.2. Evaluation Metrics

We evaluated the system across one dimension: retrieval system performance. Since we used the entire dataset for both fine-tuning and evaluation, these metrics reflect the combined performance of all components on the same more than 5000 QA pairs. Below, we provide the formulas for each metric to clarify their computation.

Retrieval Metrics:
Accuracy@k: Measures the proportion of queries where the correct document is retrieved within the top k:

$$Accuracy@k = \frac{1}{N}\sum_{i=1}^{N}\mathbf{1}\{rank(d_i^+ \mid q_i) \leq k\} \tag{5}$$

Precision@k: Assesses the fraction of retrieved documents that are relevant within the top k:

$$Precision@k = \frac{1}{k}\sum_{j=1}^{k}\mathbf{1}\{d_j isrelevant\} \tag{6}$$

Recall@k: Evaluates the fraction of relevant documents retrieved within the top k:

$$Recall@k = \frac{\sum_{j=1}^{k}\mathbf{1}\{d_j isrelevant\}}{R} \tag{7}$$

NDCG@10 (at k=10): Measures the ranking quality of retrieved documents, accounting for the position of relevant items:

$$NDCG@k = \frac{1}{IDCG@k} \sum_{j=1}^{k} \frac{2^{rel_j} - 1}{\log_2(j+1)} \tag{8}$$

MRR@10 (at k=10): Focuses on the rank of the first relevant:

$$MRR@k = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{rank_i} \tag{9}$$

MAP@100 (at k=100): Assesses overall retrieval precision across all:

$$MAP@k = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{R_i} \sum_{j=1}^{k} Precision@j \cdot \mathbf{1}\{d_j isrelevant\} \tag{10}$$

Hits@k: Proportion of times the correct answer appears in the top-k predicted results:

$$Hits@k = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\{rank(d_i^+ \mid q_i) \leq k\} \tag{11}$$

3.2. Results & Analysis

3.2.1. Halong Embedding Performance

The retrieval component of the Retrieval-Augmented Generation (RAG) system was implemented with a robust training and evaluation setup. ll experiments were conducted on Google Colab, utilizing an NVIDIA T4 GPU with 16GB VRAM and 32GB of system RAM to ensure efficient computation for the hiieu/halong_embedding model. The software environment leveraged Python 3.8 and PyTorch, optimized for deep learning tasks with the Halong Embedding model. The fine-tuning process was performed on a comprehensive dataset comprising more 5000 QA pairs sourced from youmed.vn, covering diverse medical topics such as symptoms, treatments, and disease causes. The process spanned 10 epochs, achieving a final training loss of 0.011, an average training loss of 0.07285, and a stable gradient norm of 0.13292. With a learning rate decaying to 0.0 across 460 global steps, training completed in 1378.46 seconds (approximately 23 minutes), processing 42.134 samples per second and 0.334 steps per second. Evaluation on the same dataset yielded a sequential score of 0.9154, indicating robust retrieval performance.

Table 2. Comparison of Retrieval Metrics for Halong Embedding (Dim=128) Before and After Fine-Tuning

Metric	Before Fine-Tuning	After Fine-Tuning	Improvement (%)
Accuracy@1	0.5489	0.7781	22.92
Accuracy@3	0.7994	0.9935	19.40
Accuracy@5	0.8483	0.9993	15.10
Accuracy@10	0.8920	1.0000	10.80
Precision@1	0.5489	0.7781	22.92
Precision@3	0.2665	0.3312	6.47
Precision@5	0.1697	0.1999	1.08
Precision@10	0.0892	0.1000	1.08
Recall@1	0.5489	0.7781	22.92
Recall@3	0.7994	0.9935	19.40
Recall@5	0.8483	0.9993	15.10
Recall@10	0.8920	1.0000	10.80
NDCG@10	0.7350	0.9157	18.07
MRR@10	0.6831	0.8860	20.29
MAP@100	0.6866	0.8860	19.94

The fine-tuned model demonstrated remarkable gains in retrieval accuracy and recall, reflecting a strong ability to find relevant documents. Accuracy@1 and Recall@1 both improved by 22.92% (from 0.5489 to 0.7781), indicating a significant enhancement in retrieving the single most relevant document. This is critical for precise medical Q&A, where the top result drives answer quality. At top 3, Accuracy@3 and Recall@3 increased by 19.40% (from 0.7994 to 0.9935), reflecting near-perfect retrieval, which is essential for capturing diverse medical contexts. Similar gains were seen in Accuracy@5 and Recall@5 (up 15.10%), showing robust coverage. Notably, Accuracy@10 and Recall@10 achieved a perfect score of 1.0000 (up 10.80%), ensuring comprehensive retrieval within the top 10.

While Precision improvements were less pronounced at higher ranks, the gains were significant at the top. Precision@1 matched Accuracy@1's 22.92% improvement, confirming a focus on top-1 relevance. However, Precision@3, @5, and @10 showed smaller gains of 6.47%, 3.02%, and 1.08%, respectively. This suggests the fine-tuning of prioritized ranking the most relevant documents highly, a suitable strategy for medical Q&A where top-k results are most critical. The model's superior ranking quality is also reflected in NDCG@10, which improved by 18.07% (from 0.7350 to 0.9157), ensuring the most relevant documents are prioritized for the Large Language Model. MRR@10 increased by 20.29% (from 0.6831 to 0.8860),

demonstrating an enhanced ability to place the first relevant document higher in the ranking, which is crucial for efficient retrieval. Finally, MAP@100 improved by 19.94% (from 0.6866 to 0.8860), highlighting better overall retrieval precision across a larger document set, a vital aspect of comprehensive medical

3.2.2. Keyword Search Performance(BM25)

The BM25 keyword search method, a baseline for sparse retrieval, was evaluated on more than 5000 medical QA pairs from youmed.vn, using 4 parallel threads to process 1,452 batches per thread. Each thread completed in approximately 136–140 seconds (averaging 10.34-10.63 iterations per second), totaling 141 seconds for the full evaluation. Table 3 presents the retrieval metrics for BM25, assessed across the same dataset as the Halong Embedding model.

Table 3. Retrieval Metrics for BM25 Keyword Search

Metric	**Value**	**Halong (Dim=128, Post-Fine-Tuning)**
Accuracy@1	0.4769	0.7781 (38.67%)
Accuracy@3	0.7082	0.9935 (28.68%)
Accuracy@5	0.7731	0.9993 (22.63%)
Accuracy@10	0.8278	1.0000 (17.22%)
Precision@1	0.4769	0.7781 (38.67%)
Precision@3	0.2361	0.3312 (28.72%)
Precision@5	0.1546	0.1999 (22.64%)
Precision@10	0.0828	0.1000 (17.22%)
Recall@1	0.4769	0.7781 (38.67%)
Recall@3	0.7082	0.9935 (28.68%)
Recall@5	0.7731	0.9993 (22.63%)
Recall@10	0.8278	1.0000 (17.22%)
NDCG@1	0.4769	0.7781 (38.67%)
NDCG@3	0.6158	0.9157 (32.76%)
NDCG@5	0.6426	0.9157 (29.81%)
NDCG@10	0.6006	0.9157 (27.87%)
MRR	0.6114	0.8860 (30.98%)

The BM25 evaluation demonstrates notable retrieval performance but also reveals limitations compared to more advanced embedding models. BM25 achieves an Accuracy@3 and Recall@3 of 0.7082, retrieving relevant

documents for 70.82% of queries within the top-3. This performance increases to an Accuracy@10 and Recall@10 of 0.8278, covering 82.78% of relevant documents in the top-10. However, these scores are significantly lower than Halong Embedding's post-fine-tuning scores (e.g., BM25's Accuracy@3 is 28.68% lower than Halong's 0.9935), highlighting BM25's limitation in capturing semantic relationships. In terms of Precision, Precision@3 (0.2361) and Precision@10 (0.0828) are modest, trailing Halong's scores due to BM25's reliance on exact keyword matches rather than contextual understanding. Similarly, NDCG@3 (0.6158) and NDCG@10 (0.6606) lag behind Halong, and BM25's MRR (0.6114) is 30.98% lower than Halong's, reflecting a weaker ranking of the first relevant document.

Despite its performance limitations, BM25 remains a valuable component in a RAG framework due to its computational efficiency. The BM25 evaluation completed in just 141 seconds across 4 threads, demonstrating its suitability for large-scale keyword searches. In an integrated system, BM25 can complement the semantic richness of embedding models like Halong Embedding and the structured insights of KG retrieval. By combining these complementary strengths through techniques like Reciprocal Rank Fusion, we can leverage BM25's advantage in matching exact medical terms while mitigating its semantic weaknesses. This helps provide a more accurate and comprehensive context for large language models such as Google Gemini, thereby reducing the risk of misinformation in medical Q&A.

3.2.3 KG Evaluation

To assess the quality of our constructed knowledge graph and its suitability for downstream applications such as retrieval-augmented generation (RAG), we conducted a series of experiments using knowledge graph embedding (KGE) models implemented in PyKEEN. The goal of these experiments is to evaluate how well different embedding methods capture the structural and semantic patterns in the KG, measured through the standard task of link prediction.

The dataset was randomly split into training (80%), validation (10%), and test (10%) subsets. We trained four widely used KGE models-RotatE. During evaluation, for every test triple (h,r,t) the model is asked to predict the missing head or tail entity. The predicted entities are ranked against all candidates, and performance is computed under the filtered setting, where other true triples are removed from the ranking to avoid unfair penalization. We report the following standard metrics MRR: The average of reciprocal

ranks of the correct entities. Higher values indicate better performance. MR: The average rank of the correct entities. Lower values are preferred. Hits@k (k = 1, 3, 10): The proportion of correct entities ranked in the top-k positions. Higher values reflect stronger ranking ability.

Table 4. Link Prediction Evaluation Performance

Metric	**MRR**	**Hits@1**	**Hits@3**	**Hits@10**
Value	0.229	0.214	0.227	0.254

4. Conclusion & Future Work

HealthRAG is a Vietnamese medical question-answering system that leverages a hybrid retrieval strategy, combining Halong Embedding, BM25, and Knowledge Graph to provide accurate and reliable responses. This approach demonstrates strong effectiveness in addressing both linguistic and domain-specific challenges in the Vietnamese medical context while ensuring scalability through efficient processing. By integrating multiple retrieval methods, HealthRAG enhances the reliability of medical information delivery and supports more trustworthy healthcare knowledge access.

For future work, several directions can further improve HealthRAG. Enhancing multimodal capabilities by incorporating annotated medical images will improve user comprehension. Time-aware updates can help maintain an up-to-date medical corpus, while personalization based on user profiles may deliver more contextually relevant responses. Advanced reasoning can be explored through Graph Neural Networks, and large-scale deployment with user feedback will validate the system in real-world scenarios. Optimizing efficiency with lower-dimensional embeddings and cloud-based APIs will strengthen scalability and accessibility. Finally, reinforcing safety mechanisms—such as hallucination detection and ethical compliance—will be essential to ensure trustworthy and responsible use in medical applications.

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